

Fig. 10: Tight layout to minimise power loop length

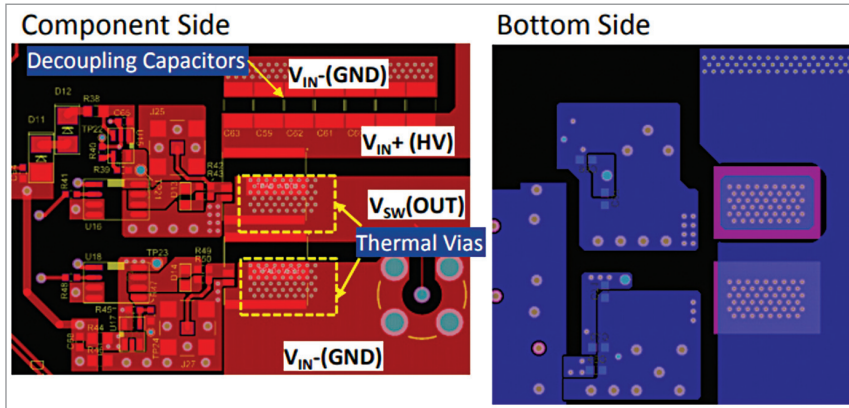


Fig. 11: Designing for low thermal resistance

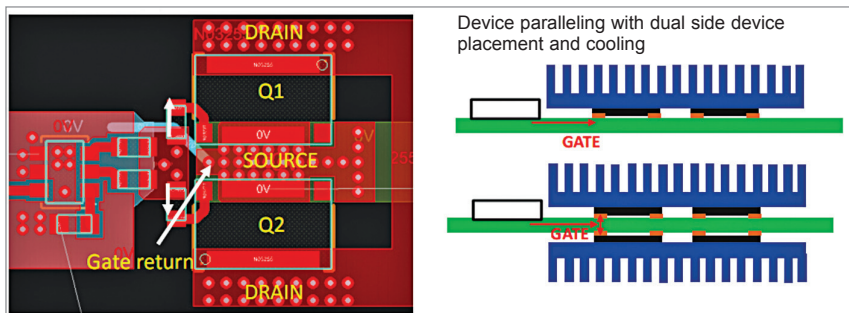


Fig. 12: Dual side placement with optimum layout

which effectively appears as voltage drop plus a drop proportional to $1/\text{transconductance}$.

In power converters, there is a short time in each cycle—the dead time—where neither switch is on, and the inductor current is forced to flow through one of the FETs. The energy lost in the dead time is proportional to the current, voltage, and the amount of time spent in this mode, which corresponds directly to the dead time.

It can be easily deduced that the dead time must be minimised for low power loss. But smaller dead time will result in the loss of zero voltage switching. If the dead time goes negative, however, cross conduction will occur with the associated loss in efficiency. Therefore, the dead time

must be held to a small band without straying outside it.

Smaller dead time and negative voltage driving can increase the efficiency of the power converters significantly. Nowadays, gate drivers come with programmable dead time. When determining what dead time is sufficient for a circuit, various mismatches and effects of propagation delay must be taken into consideration for precise control. Following are those aspects:

- (i) Mismatch of the driver. For drivers with separate high and low side inputs, it is important to look for the difference in propagation delays
- (ii) Propagation delay variation with high-side ground
- (iii) Variation with bootstrap voltage

(iv) When the signal is propagating through the level shifter simultaneously with a switching transition

5. Crosstalk suppression. When GaN devices work at high frequencies, the high dv/dt and di/dt due to parasitic inductances and capacitances leads to oscillations and overshoots in gate-source voltage and affect the stability of driving circuits. Moreover, it gives rise to crosstalk between the upper and lower switch in certain topologies.

The circuit in Fig. 5 can effectively suppress crosstalk as it can maintain an appropriate negative voltage when the gate positive voltage peak occurs and pulls up the gate voltage to a less negative voltage level or zero level before the gate negative overshoot event occurs. This circuit raises the gate voltage to 0V and clamps it at 0V after the false triggering period, and before the turn-on signal arrives. This makes the negative gate overshoot occur at 0V level, reducing the risk of negative breakdown. Moreover, a low turn-off impedance path is created to suppress crosstalk, which can limit the positive and negative false triggering voltage peaks to the required range because the common source inductor (CSI) is decoupled and displacement current of the gate-drain capacitor is bypassed.

6. Adding ferrite beads or RC snubbers. Ferrite beads are well known for suppressing high-frequency electronic noise in electronic circuits. They employ high-frequency current dissipation in a ferrite ceramic to build high-frequency noise suppression devices. It is a common method of inserting ferrite beads into the power loop to dampen parasitic ringing.

The equivalent circuit of the ferrite bead is a series connection of frequency-dependent resistor and inductor. Therefore, the ferrite bead and the device parasitic capacitance form a second-order circuit with oscillation frequency expressed as $f_{osc} = 1/2\pi\sqrt{C_{eq}L_{eq}}$. Here, C_{eq} is the equivalent parasitic capacitance of the device, and L_{eq} is the equivalent parasitic inductance of the circuit and the ferrite bead.

The damping coefficient is given by:

$$m = R_{eq}/2\sqrt{L_{eq}/C_{eq}}$$

where R_{eq} is the resistive part of the